

Name:, NEPTUN-code

Group, teacher:

Computer Science BSc Analysis-1 TEST-1
25-th of March, 2025

Reasoning and justification are needed in the solutions.

1. (9 points) Determine (without using the concept of limit) $\sup H$, $\inf H$, $\max H$, $\min H$ if

$$H = \left\{ \frac{3\sqrt{x} + 5}{2\sqrt{x} + 3} \mid x \in [4, +\infty) \right\}.$$

2. (8 points) Evaluate the following limit, then prove BY DEFINITION your result:

$$\lim_{n \rightarrow \infty} \frac{3n^3 - n^2 + 12n - 10}{n^3 + n^2 + 3n + 5}$$

3. Evaluate the following limits:

(a) (6 points) $\lim_{n \rightarrow \infty} (\sqrt{9n^2 + 3n - 1} - 3 \cdot \sqrt{n^2 - n + 1}) = ?$

(b) (6 points) $\lim_{n \rightarrow \infty} \sqrt[n]{\frac{3^{2n+1} + n^3 \cdot 2^{n+1}}{4^n + 9^{n-1}}} = ?$

(c) (6 points) $\lim_{n \rightarrow \infty} \left(\frac{2n+5}{2n-1} \right)^{n-1} = ?$

4. (8 points) Prove that the following sequence is convergent, and compute its limit:

$$a_1 = \frac{1}{2}, \quad a_{n+1} = \frac{1}{2 - a_n} \quad (n \in \mathbb{N}).$$

5. (7 points) Consider the following function:

$$f(x) := x^2 - 2x \quad (x \in [2; +\infty)).$$

Prove that f is invertible and give the inverse function of f (domain, formula and range of values of the inverse function).